# Paul Parisot

Software Engineer - C++ & Low-Latency Systems

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## About

Systems software engineer deeply focused on modern C++ performance optimization, low-level Linux runtimes, and hardware-software boundaries.
Specialized in zero-allocation memory design, concurrent thread synchronization, and asynchronous networking architectures.
Pragmatic debugger experienced in diagnosing subtle performance bottlenecks, memory corruptions, and race conditions in production environments.

## Skills

**Core Systems**: C++ (Modern C++17/20), C, Go, Python, x86/ARM Assembly stubs
**Low-Level Linux**: Memory Management (`heap, stack, shm`), POSIX Threads, Atomics, Driver Communication Boundaries
**Performance & Networking**: Asynchronous I/O (`Boost.Asio`), TCP/IP Paradigms, Cache-Line Linearity (ECS Layout)
**Engineering Toolchain**: Linux, CMake, GCC, GDB, Valgrind (Memcheck/Helgrind), Git, Jenkins CI/CD pipelines

## Professional Experience

**Faurecia / Forvia - Software Engineer (Systems & Tooling)**

- Developed and optimized low-level C++ software modules tailored for hardware driver communication interfaces.
- Engineered internal testing & visual tracking applications using C# (Unity engine) under strict system resource limitations.
- Analyzed physical architecture constraints to maintain deterministic software behavior and memory guarantees.
- Collaborated closely within an industrial Agile/Scrum framework, tracking requirements and features via **Jira**.

**Gno.land - Core Software Engineer (Distributed Runtime Engines)**

- Refactored core multi-threaded execution sub-systems to guarantee operational determinism and maximum execution speed.
- Isolated, reproduced, and patched **6 critical bugs** causing runtime crashes, delivering 10k+ LOC to production.
- Authored automated regression and integration testing suites embedded into continuous deployment (CI/CD) workflows.

**Epitech / IONIS - Systems Programming Instructor & Code Auditor**

- Taught advanced C/C++ memory architectures, pointer arithmetic, and native Linux system programming to **400+ students**.
- Audited large-scale repositories to enforce strict production standards: error-handling paths, cache awareness, and memory safety.

## Low-Latency & Systems Technical Projects

**rodox – Low-Latency Linux Systems Engine**

Built a modular runtime infrastructure in C++ using a data-oriented Entity Component System (ECS) to avoid cache misses.
Completely bypassed OS-level heap allocation latencies (`malloc`/`new`) by designing thread-safe **custom static pool allocators**.
Implemented high-throughput, non-blocking asynchronous networking components utilizing `Boost.Asio` over TCP. Hardened execution paths against hardware-software latency spikes using profiling tools including **CMake**, **GDB**, and **Valgrind**.

**Skills:** Modern C++, SIMD, Boost.Asio, Memory Management, Linux System APIs, Profiling (Valgrind), Concurrency

**cclass – Runtime Memory Mutation Engine**

Engineered a raw C engine capable of dynamically rewriting compiled assembly instructions inside secure execution memory regions.
Leveraged system-level `mmap` calls with explicit page protections to remove virtual dispatch runtime overheads.
Enforced strict memory alignment checks to prevent hardware execution faults across target host platforms.

**Skills:** C, Linux Systems, x86 Assembly, mmap, Memory Safety, Low-level Debugging

**Sledge – Containerized Orchestration & Core Backend**

Architected and implemented a high-performance core backend infrastructure in Go optimized for high-volume execution streams.
Utilized the Docker API at runtime to dynamically spin up, isolate, and orchestrate concurrent environment containers.
Wrote automated validation and monitoring tooling using **Python** scripts to verify continuous data flow and system state health.

**Skills:** Go, Docker API, Python Scripting, Data Streaming, System Automation

## Education

**Epitech, Paris** 2022 – 2027
Master's Degree in Information Technology (Master's level diploma)
Level-7 Certification - Expert in Software Engineering
Specialization: Systems Architecture

**McGill University, Montréal** 2025 – 2026
Certificate in Management
Technical Leadership & Strategy